

WISCONSIN MULTIFAMILY BENCHMARK

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**BY SCOTT PIGG AND
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A building owner or operator asks: How does the energy bill for my 1970s era, 20-unit apartment building compare to the energy bills of other apartment buildings in the city? A developer asks: Should we install individual furnaces, a condensing boiler, or a closed water loop heat pump system in our new multifamily property? The staff of Madison Gas and Electric (MGE)—the primary energy supplier for the growing metropolitan area of Madison, Wisconsin—gets these kinds of questions often. MGE decided in 2005 to undertake a study of multifamily buildings in its service territory, and to build usable benchmarks for multifamily building energy costs so that we could answer these questions accurately.

But comparing heating and cooling options for multifamily buildings is complicated. Different heating and cooling strategies involve different fuels. And the fuel may be paid for by tenants, or by landlords, or even through commercial tenant utility accounts. So MGE's goal was simply to give its customers a sense of the range of electricity and natural gas costs across a variety of larger multifamily properties. To conduct the study, MGE enlisted the assistance of the Energy Center of Wisconsin. Dalhoff and Associates of Madison, Wisconsin, also provided assistance with on-site data collection.



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Choosing Our Buildings

Our approach to the study was simple: Analyze a sample of multifamily properties by conducting brief on-site verification of dwelling units, floor area, and mechanical equipment, and analyze a year's worth of monthly utility energy cost and usage data. MGE staff recruited sites for the study from among the multifamily property operators in MGE's service territory. Additional sites came from properties randomly sampled as part of the Energy Center's recent Rental Characterization Study (available at www.ecw.org).

We required that properties in the study have eight or more units. We attempted to balance the study group across a number of different heating system types. In this, we were only partly successful; most of the buildings had either boilers with hydronic distribution, closed water loop heat pump (CWLHP) systems, or individual forced-air furnaces. But we were able to include some sites with packaged terminal air conditioning (PTAC) with either electric-resistance or heat pump space heating, some sites with baseboard electric heating, and a couple of sites that used steam systems for heating.

Figure 2. Annual Costs by Site, Sorted by Heating System Type, Decade of Construction, and Number of Stories

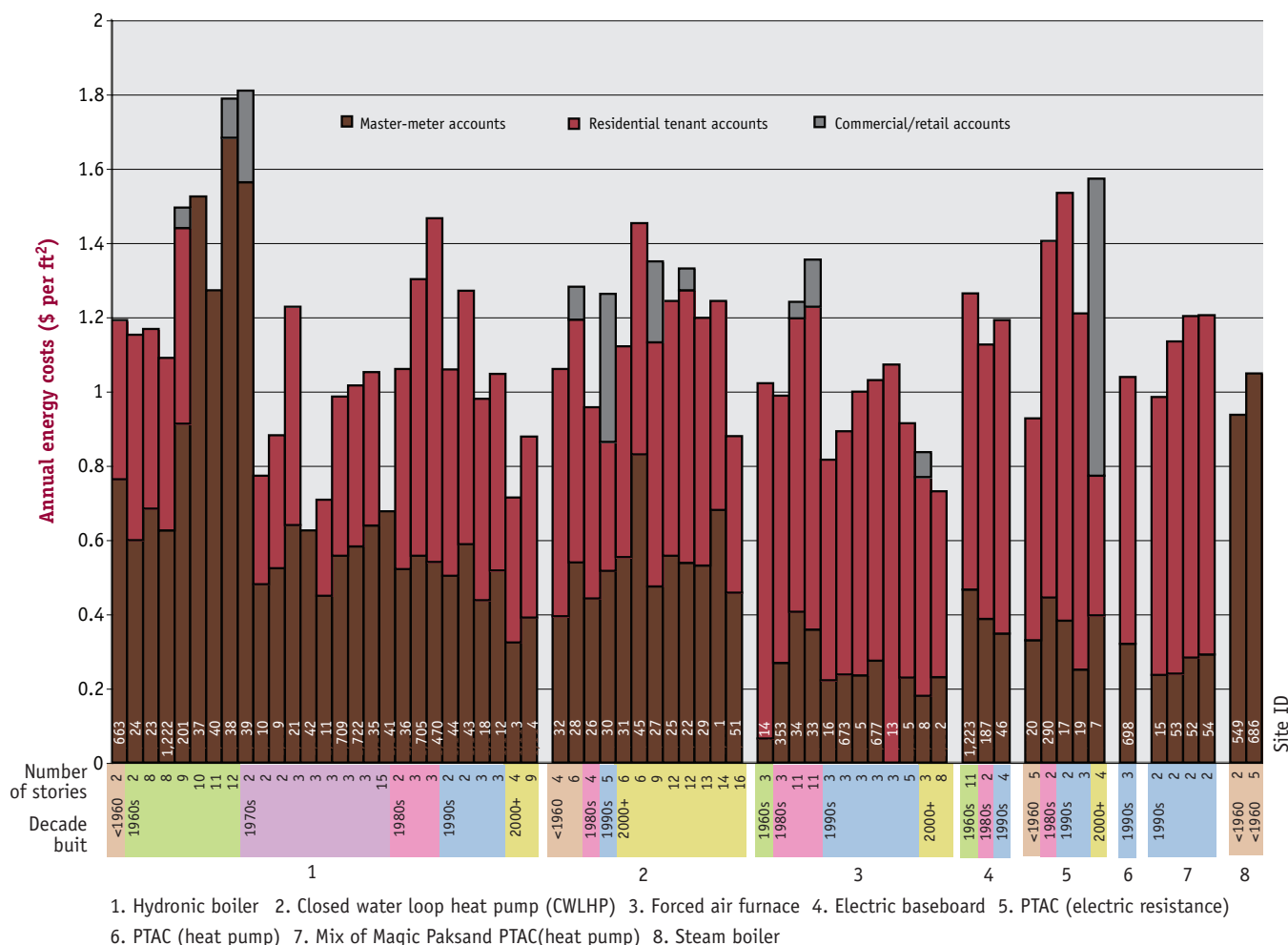


Figure 1. Our approach to the study was simple: Analyze a sample of multifamily properties by conducting brief on-site verification of dwelling units, floor area, and mechanical equipment, and analyze a year's worth of monthly utility energy cost and usage data.

The properties also varied in size and age, ranging from 8 dwelling units to more than 250, though the majority fell between 20 and 100 dwelling units. About 60% of the study group buildings were low-rise (with four or fewer stories). About half of the group could be considered new (built on or after 1990). The newer buildings were more likely to be mixed use, with commercial or retail tenants in addition to residential tenants.

Building a Benchmark

We looked at total electricity and natural gas costs/ft² from September 2004 to September 2005, including all account types: tenant, landlord, and

commercial. The results indicate a threefold range in total energy costs/ft² of floor space, from \$0.62/ft² to \$1.81/ft² (see Figure 1). The median building had an annual energy cost of \$1.10/ft², and half of the properties fell between \$0.98/ft² and \$1.26/ft². During this period, the average retail cost of electricity and natural gas (including fixed meter charges) was about \$0.13/kWh and \$1.43/therm for tenant accounts, and about \$0.08/kWh and \$0.87/therm for master meter accounts—though individual buildings were on a variety of MGE rate schedules.

Hydronically heated buildings had both the highest and the lowest energy costs/ft² in the study. This finding reflects the diversity of these buildings,

which ranged from older high-rise properties to a new low-rise property with high-efficiency condensing boilers, radiant in-floor distribution, and solar-assisted domestic hot water.

The 12 buildings with forced-air systems (of which 9 had noncondensing furnaces and 3 had condensing furnaces) had the lowest median operating cost; but building size and age are important confounding factors that cloud this simple comparison, and this group was dominated by relatively new, low-rise properties.

On average, the CWLHP buildings had somewhat higher operating costs than buildings with hydronic distribution or forced-air systems, even though this group included more new buildings.

But the CWLHP group also had more high-rise buildings, and had higher non-space-conditioning electric loads, such as electric clothes dryers, refrigerators, lighting, computers, and other electronics with chargers. It is not clear why non-space-conditioning use is higher in CWLHP buildings. Perhaps they have a greater density of people, electric end uses, or both. At least half of CWLHP buildings house student populations. Several of the remaining buildings are luxury condominiums, where household electric end uses also abound. (For more on the plug load problem, see “Roadblocks to Zero-Energy Homes,” *HE* Jan/Feb '07, p. 24.)

The on-site data gathered as part of this project—as well as common sense—suggest that because tall buildings cover more floors with a single roof area, they have less exposed surface area/ft² of floor area than short buildings. On the other hand, tall buildings are more exposed to wind and have significantly higher stack effect pressures that could lead to greater infiltration losses. Indeed the data suggest that high-rise buildings have higher average operating costs than low-rise buildings. In any event, it is plausible that differences in building height (and age) could account for some of the observed differences across heating system types. It is also plausible that tenant demographics or other nonbuilding factors could contribute to differences in energy costs across groups.

We also estimated space-heating and -cooling costs and compared these across the buildings in the sample. To do so, we calculated the best-fit heating and cooling slopes (consumption/°F) for each building using Madison weather data, and then used this information to infer annual heating and cooling

consumption in a typical year. Because this procedure relies on statistically estimating the relationship between energy consumption and outdoor temperature to separate heating and cooling energy use from other end uses, the estimates that we derived are only approximate. In addition, several buildings for which we had insufficient data, or that showed a poor correlation between usage and temperature, were omitted from this analysis.

electricity consumption for uses other than space conditioning. We expect that this could reduce heating loads in these buildings and increase cooling loads.

It is also informative to look at the most energy-efficient building in each of the three main categories of heating system (see Figure 2). All three of these buildings are relatively new, and each one might be viewed as a state-of-the-art building in its category. Of these buildings, which range from 4 to 13 stories, the ones with hydronic and forced-air systems showed comparable heating and cooling costs, while the (taller) CWLHP building showed significantly higher heating costs but somewhat lower cooling costs than the other two.

All of the preceding analyses include electricity and natural gas rates as a potential confounding factor. Bear in mind, however, that rates may partially depend on the type of HVAC system installed in the building. Central equipment is more likely to be on a commercial account paid by a landlord, while individual

equipment is more likely to be billed directly to the tenants at residential rates. To correct for this effect, we calculated heating and cooling costs using standardized average rates across all buildings in the study (\$0.958/therm for natural gas, \$0.1087/kWh for space-heating electricity, and \$0.1307/kWh for space-cooling electricity). To further control for differences across buildings while still allowing visual examination of the data, we calculated these costs/ft² of exposed shell area. From this perspective—and focusing on the three commonest types of heating system—CWLHP buildings had the highest average space-conditioning costs (a median of \$0.55/ft² of

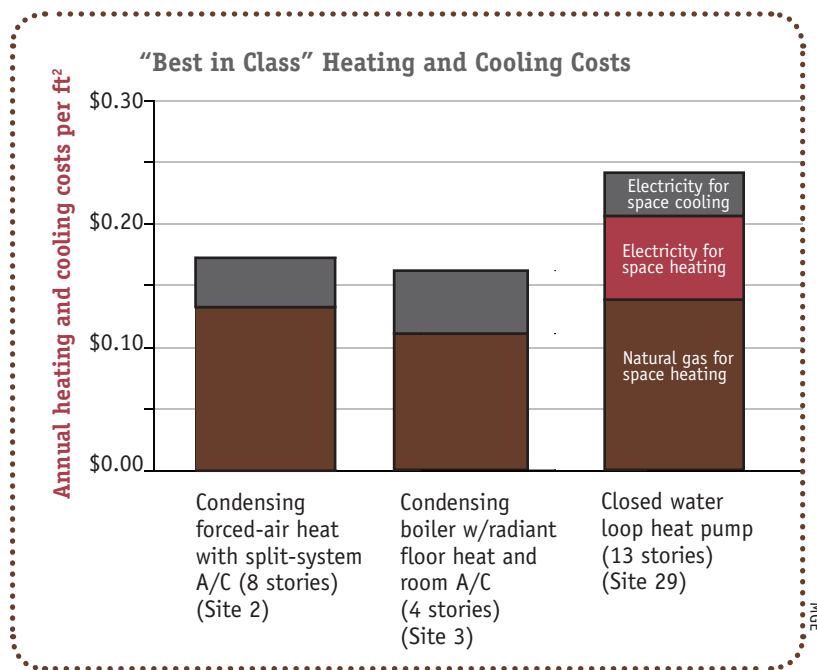


Figure 2. It is also informative to look at the most energy-efficient building in each of the three main categories of heating system. All three of these buildings are relatively new, and each one might be viewed as a state-of-the-art building in its category.

We found that heating costs had a greater (fivefold) range than total energy costs—from \$0.11/ft² to \$0.62/ft². There is great potential here for improvement, especially in newly designed buildings. When we analyzed heating and cooling costs by heating system type, we found that the CWLHP buildings averaged somewhat lower heating costs/ft² than buildings with either hydronic or forced-air systems, but the CWLHP buildings also showed higher cooling costs. We do not know whether this is due to intrinsic differences among the heating systems themselves, or to other confounding factors. As a group, the CWLHP buildings had higher estimated

shell area), followed by hydronic boiler systems (a median of \$0.39/ft² of shell area), and forced-air systems (a median of \$0.31/ft² of shell area).

It is noteworthy that the 7 buildings with electric-resistance heat (baseboard or resistance PTAC) had about half the normalized heating use, on average, of the 17 buildings in the Energy Center's Rental Characterization Study, even after we allow for regional differences in heating degree-days. The electric-resistance buildings in this study may not be representative of the broader population of electrically heated buildings, however.

Finally, to simultaneously control for the various confounding factors, we did a regression analysis of rate-standardized heating and cooling costs/ft² of floor area as a function of heating system type, building height, building age, and non-space-conditioning electricity consumption. The results suggest only slightly higher energy costs for

CWLHP systems compared to hydronic heating (the difference is not statistically significant). Forced-air systems, on the other hand, exhibited statistically significantly lower costs—though the size of this difference was not very precisely determined from the regression.

Help for Multifamily Building Stakeholders

Overall, MGE's benchmarking data should help building owners, developers, and operators to understand typical heating and cooling costs for various types of multifamily building in MGE's service territory. In particular, these data will allow for a more factual discussion of HVAC options when developers and owners are still in the planning and design stage, allow them to prepare a more accurate energy budget for a particular project, and make better local energy decisions as these buildings are built or rehabbed.

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For more information:

To learn more about the MGE study, go to www.mge.com/business/saving/multifamily.htm.

The results of this study are also compiled on Oak Ridge National Laboratory's Web site on benchmarking building energy performance, at <http://eber.ed.ornl.gov/benchmark/wisc.htm>. This Web site allows owners and operators of multifamily buildings to compare their buildings' performance to the MGE study findings.

